

Sample Test 1: Logical Proof

Nguyen Dinh Thien Loc

Student ID: 24125093

April 19, 2026

Problem 2

(a)

1.	$\exists x(P(x) \wedge Q(x))$	premise
2.	$\forall x(Q(x) \rightarrow R(x))$	premise
3.	$x_0 \quad P(x_0) \wedge Q(x_0)$	assumption
4.	$Q(x_0) \rightarrow R(x_0)$	$\forall x e \ 2$
5.	$Q(x_0)$	$\wedge e_2 \ 3$
6.	$R(x_0)$	$\rightarrow e \ 4, 5$
7.	$P(x_0)$	$\wedge e_1 \ 3$
8.	$P(x_0) \wedge R(x_0)$	$\wedge i \ 7, 6$
9.	$\exists x(P(x) \wedge R(x))$	$\exists x i \ 8$
10.	$\exists x(P(x) \wedge R(x))$	$\exists x e \ 1, 3-9$

(b)

1.	$\exists xP(x)$	premise
2.	$\forall x\forall y(P(x) \rightarrow Q(y))$	premise
3.	y_0	
4.	$x_0 \quad P(x_0)$	assumption
5.	$\forall y(P(x_0) \rightarrow Q(y))$	$\forall x e \ 2$
6.	$P(x_0) \rightarrow Q(y_0)$	$\forall y e \ 5$
7.	$Q(y_0)$	$\rightarrow e \ 6, 4$
8.	$Q(y_0)$	$\exists x e \ 1, 4-7$
9.	$\forall yQ(y)$	$\forall y i \ 3-8$

Problem 3

(a) $\forall x(P(x) \rightarrow Q(f(x)))$

Model $\mathcal{M}_1$ (Satisfies the formula)

- **Domain** $\mathcal{D} = \{1, 2\}$
- **Function** $f(n) = n$ (Identity function)
- **Predicates:** $P = \{1\}, Q = \{1, 2\}$
- **Reason:** For $x = 1$, $P(1) \rightarrow Q(1)$ is $T \rightarrow T$ (True). For $x = 2$, $P(2) \rightarrow Q(2)$ is $F \rightarrow T$ (True).

Model $\mathcal{M}_2$ (Does not satisfy)

- **Domain** $\mathcal{D} = \{1\}$
- **Function** $f(1) = 1$
- **Predicates:** $P = \{1\}, Q = \emptyset$
- **Reason:** For $x = 1$, $P(1) \rightarrow Q(1)$ becomes $T \rightarrow F$, which is False.

(b) $\exists x \forall y ((P(x, y) \wedge P(y, x)) \vee (Q(x, y) \wedge R(y, x)))$

Model $\mathcal{M}_3$ (Satisfies the formula)

- **Domain** $\mathcal{D} = \{a\}$
- **Predicates:** $P = \{(a, a)\}, Q = \emptyset, R = \emptyset$
- **Reason:** If we pick $x = a$, then for all y (which can only be a), the first part of the disjunction $(P(a, a) \wedge P(a, a))$ is True. Thus the whole formula is True.

Model $\mathcal{M}_4$ (Does not satisfy)

- **Domain** $\mathcal{D} = \{1, 2\}$
- **Predicates:** $P = \emptyset, Q = \emptyset, R = \emptyset$
- **Reason:** All predicates are empty relations. For any x we choose, both $(P \wedge P)$ and $(Q \wedge R)$ will be False for all y , making the entire disjunction False.

Problem 4

(a) $\exists x P(x) \wedge \exists x Q(x) \models \exists x (P(x) \wedge Q(x))$

Result: This semantic entailment **does not hold**.

Counter-example Model

- **Domain** $\mathcal{D} = \{1, 2\}$
- **Predicates:** $P = \{1\}$, $Q = \{2\}$
- **Reason:**
 - $\exists x P(x)$ is True (witness $x = 1$).
 - $\exists x Q(x)$ is True (witness $x = 2$).
 - Thus, the premise $\exists x P(x) \wedge \exists x Q(x)$ is True.
 - However, there is no element in $\mathcal{D}$ that is in both P and Q . Thus, $\exists x (P(x) \wedge Q(x))$ is False.

(b) $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$

Result: This semantic entailment **holds**.

Proof Sketch

Assume the premise $\exists y \forall x P(x, y)$ is true in a model $\mathcal{M}$ with environment l . This means there exists some element $c \in \mathcal{D}$ such that for all elements $a \in \mathcal{D}$, the pair $(a, c) \in P^{\mathcal{M}}$. To show $\forall x \exists y P(x, y)$, we must show that for any arbitrary $a \in \mathcal{D}$, there exists some $b \in \mathcal{D}$ such that $(a, b) \in P^{\mathcal{M}}$. For any such a , we can choose $b = c$. Since the premise guarantees $(a, c) \in P^{\mathcal{M}}$ for all a , the conclusion is satisfied.

Problem 5

(a) $\phi = ((p \rightarrow q) \wedge p) \vee q$

To check for validity, we apply the Tableau Algorithm to the negation $\neg\phi$:

$$\neg(((p \rightarrow q) \wedge p) \vee q)$$

Tableau Path for $\neg\phi$

1. $C_0 = \{\neg(((p \rightarrow q) \wedge p) \vee q)\}$
2. $C_1 = \{\neg((p \rightarrow q) \wedge p), \neg q\}$ ($\neg\vee$ -rule)
3. **Path Branching** ($\neg\wedge$ -rule applied to $\neg((p \rightarrow q) \wedge p)$):
 - **Branch 1:** $C_2 = \{\neg(p \rightarrow q), \neg q\}$
 - $C_3 = \{p, \neg q, \neg q\}$ ($\neg\rightarrow$ -rule)
 - This path is **complete** and **not closed** (no clash).
 - **Branch 2:** $C_2 = \{\neg p, \neg q\}$
 - This path is **complete** and **not closed** (no clash).

Conclusion: Since there exists a complete tableau path that is not closed, the negated formula $\neg\phi$ is **satisfiable**[cite: 1877]. Therefore, the original formula ϕ is **not valid**.

(b) $\psi = \forall x\neg P(x) \vee \exists x(P(x) \vee Q(x))$

We apply the Tableau Algorithm to the negation $\neg\psi$:

$$\neg(\forall x\neg P(x) \vee \exists x(P(x) \vee Q(x)))$$

Tableau Path for $\neg\psi$

1. $C_0 = \{\neg(\forall x\neg P(x) \vee \exists x(P(x) \vee Q(x)))\}$
2. $C_1 = \{\neg\forall x\neg P(x), \neg\exists x(P(x) \vee Q(x))\}$ ($\neg\vee$ -rule)
3. $C_2 = \{\exists x\neg\neg P(x), \neg\exists x(P(x) \vee Q(x))\}$ ($\neg\forall$ -rule)
4. $C_3 = \{\exists xP(x), \neg\exists x(P(x) \vee Q(x))\}$ ($\neg\neg$ -rule)
5. $C_4 = \{\exists xP(x), \forall x\neg(P(x) \vee Q(x))\}$ ($\neg\exists$ -rule)
6. $C_5 = \{P(c), \forall x\neg(P(x) \vee Q(x))\}$ ($\exists$ -rule, creates new constant c)
7. $C_6 = \{P(c), \neg(P(c) \vee Q(c))\}$ ($\forall$ -rule using c)
8. $C_7 = \{P(c), \neg P(c), \neg Q(c)\}$ ($\neg\vee$ -rule)

Conclusion: The path C_7 contains a **clash** $\{P(c), \neg P(c)\}$ [cite: 2057]. Since all possible fair tableau paths are closed, the negated formula $\neg\psi$ is **unsatisfiable**[cite: 2121]. Therefore, the original formula ψ is **valid** (a tautology)[cite: 2124].